THE PROJECTION FORMULA FOR CAPPING WITH A CHERN CLASS AND PROPER PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The characterisation of the Chern classes by one relation on the projective bundle. Same chapter chow.tex, label lemma-determine-intersections, invoked at proof lines 7412 and 7435. Gives the relation satisfied by the classes obtained by capping with the Chern classes of the pulled back sheaf, and then the uniqueness that identifies the pushed forward classes as the Chern classes of the sheaf on the target.
- Compatibility of the cap with a first Chern class and flat pullback, with proper pushforward, and preservation of rational equivalence. Same chapter chow.tex, labels lemma-pushforward-cap-c1 (proof line 7428), lemma-flat-pullback-proper-pushforward (proof line 7429) and lemma-proper-pushforward-rational-equivalence (proof line 7426). They let the relation be pushed forward along the morphism between the projective bundles and rewritten in terms of the pushed forward classes, which is where the projection formula is used.

Source reference. Original lemma 02U6: sources/stacks-chow/pinned-original-source.tex, source lines 7280–7280. Navigation pointer for \ref{lemma-determine-intersections} used inside the counted span (line 7412); the argument that consumes it is retained in the span.

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Source reference. Original lemma 02RG: sources/stacks-chow/pinned-original-source.tex, source lines 2401–2401. Navigation pointer for \ref{lemma-flat-pullback-proper-pushforward} used inside the counted span (line 7429); the argument that consumes it is retained in the span.

Source reference. Original lemma 02S2: sources/stacks-chow/pinned-original-source.tex, source lines 3217–3217. Navigation pointer for \ref{lemma-proper-pushforward-rational-equivalence} used inside the counted span (line 7426); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SU: sources/stacks-chow/pinned-original-source.tex, source lines 4631–4631. Navigation pointer for \ref{lemma-pushforward-cap-c1} used inside the counted span (line 7428); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 7395); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y be locally of finite type over S. Let $\mathcal{E}$ be a finite locally free sheaf of rank r on X. Let $p : X \to Y$ be a proper morphism. Let α be a k-cycle on X. Let $\mathcal{E}$ be a finite locally free sheaf on Y. Then*

$$p_*(c_j(p^*\mathcal{E}) \cap \alpha) = c_j(\mathcal{E}) \cap p_*\alpha$$

Proof. Let $(\pi : P \to Y, \mathcal{O}_P(1))$ be the projective bundle associated to $\mathcal{E}$. Then $P_X = X \times_Y P$ is the projective bundle associated to $p^*\mathcal{E}$ and $\mathcal{O}_{P_X}(1)$ is the pullback of $\mathcal{O}_P(1)$. Write $\alpha_j = c_j(p^*\mathcal{E}) \cap \alpha$, so $\alpha_0 = \alpha$. By Lemma 02U6 we have

$$\sum\nolimits_{i=0}^{r} (-1)^i c_1(\mathcal{O}_P(1))^i \cap \pi_X^*(\alpha_{r-i}) = 0$$

in the chow group of P_X. Consider the fibre product diagram

$$\begin{array}{ccc} P_X & \xrightarrow{p'} & P \\ {\scriptstyle \pi_X}\downarrow & & \downarrow{\scriptstyle \pi} \\ X & \xrightarrow{p} & Y \end{array}$$

Apply proper pushforward p'_* (Lemma 02S2) to the displayed equality above. Using Lemmas 02SU and 02RG we obtain

$$\sum\nolimits_{i=0}^{r} (-1)^i c_1(\mathcal{O}_P(1))^i \cap \pi^*(p_*\alpha_{r-i}) = 0$$

in the chow group of P. By the characterization of Lemma 02U6 we conclude. $\square$